

# Lagrange derivatives in MOCCa

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## I. INTRODUCTION

We will be building a Lagrange mesh in a single Cartesian direction with an **even** number  $2N$  of mesh points that are evenly spaced with a spacing  $dx$ . The points are then found at the following coordinates

$$-(N - \frac{1}{2})dx, -(N - 1 - \frac{1}{2})dx, \dots, -\frac{1}{2}dx, +\frac{1}{2}dx, \dots, (N - \frac{1}{2})dx. \quad (1)$$

Note that the boundaries of the numerical simulation volume are at  $[-Ndx, +Ndx]$  and the discretisation points find themselves in the middle of the 'cells'.

Let's first label these points as

$$x_i = -\left(N - \frac{1}{2} + i\right)dx \quad \text{for } i = 1, \dots, 2N. \quad (2)$$

Let us now fix a function  $h$  on the Lagrange mesh, characterised by its values on the mesh points  $h(x_i)$ . Expanding  $h(x)$  in the Lagrange basis functions, we can calculate the value of  $\partial_x h(x)$  and  $\partial_{xx}^2 h(x)$  at a mesh point  $x_i$  from the values of  $h$  at *all of the mesh points*: [?] ]

$$\frac{dh}{dx}(x_i) = \sum_{j=1}^{2N} D_{ij}^{(1)} h(x_j), \quad (3)$$

$$\frac{dh}{dx}(x_i) = \sum_{j=1}^{2N} D_{ij}^{(2)} h(x_j), \quad (4)$$

$$(5)$$

where the matrices  $D^{(1)}$  and  $D^{(2)}$  are the Lagrange derivatives; they each have dimension  $2N \times 2N$ . The relevant expressions are

$$D_{ij}^{(1)} = \begin{cases} (-1)^{(i-j)} \frac{\pi}{2Ndx} \frac{1}{\sin[\pi(i-j)/(2N)]} & \text{if } i \neq j \\ 0 & \text{if } i = j \end{cases}, \quad (6)$$

$$D_{ij}^{(2)} = \begin{cases} (-1)^{i-j+1} 2 \frac{\pi}{2Ndx} \frac{2 \cos[\pi(i-j)/(2N)]}{\sin^2[\pi(i-j)/(2N)]} & \text{if } i \neq j \\ -\frac{\pi^2}{3dx^2} \left(1 - \frac{1}{4N^2}\right) & \text{if } i = j \end{cases}, \quad (7)$$

where we note the correction of the misprint in Ref. [2], where Eq. (18) is affected by a spurious minus sign as pointed out by E. Tijskens.

## II. AXIS REDUCTION

Things get interesting when we now suppose that all functions  $h(x)$  that we will represent on the mesh are either symmetric or skew symmetric w.r.t. to the origin, either  $h(-x) = \pm h(x)$ . This is obviously interesting to reduce the memory required to represent  $h$  itself, but it also makes taking derivatives much less costly.

In particular, for a value of  $i > N$  and hence  $x_i > 0$ :

$$\begin{aligned}
\frac{dh}{dx}(x_i) &= \sum_{j=1}^N D_{ij}^{(1)} h(x_j) + \sum_{j=N+1}^{2N} D_{ij}^{(1)} h(x_j) \\
&= \pm \sum_{j=1}^N D_{ij}^{(1)} h(-x_j) + \sum_{j=N+1}^{2N} D_{ij}^{(1)} h(x_j) \\
&= \pm \sum_{j=1}^N D_{ij}^{(1)} h(+x_{2N-j+1}) + \sum_{j=N+1}^{2N} D_{ij}^{(1)} h(x_j) \\
&= \pm \sum_{\ell=N}^{2N} D_{i(2N-\ell+1)}^{(1)} h(x_\ell) + \sum_{j=N+1}^{2N} D_{ij}^{(1)} h(x_j) \\
&= \sum_{j=N+1}^{2N} \left[ D_{ij}^{(1)} \pm D_{i(2N-j+1)}^{(1)} \right] h(x_j) \\
&= \sum_{j=N+1}^{2N} D_{(i-N), (j-N)}^{(1), \pm} h(x_j) \\
&= \sum_{j=1}^N D_{(i-N), j}^{(1), \pm} h(x_{j+N})
\end{aligned} \tag{8}$$

where we have made the substitution  $\ell = 2N - j + 1$  (and hence  $j = 2N - \ell + 1$ ) in the first sum to go from the third to the fourth line. Eq. (8) means that we can calculate the derivative of  $h$  on the positive part of the mesh with explicit storage only of  $h$  on that same part of the mesh, provided we use the derivative matrix  $D^{(1), \pm}$  for the symmetry of the function  $h$ . To be explicit about their definition, the matrices  $D^{(1), \pm}$  have definition for

$$D_{ij}^{(1), \pm} = D_{ij}^{(1)} \pm D_{i(N-j+1)}^{(1)} \tag{9}$$

where we emphasize that both  $i$  and  $j$  can only range from 1 to  $N$  in this equation! The matrices  $D^{(1), \pm}$  are what the current Fortran implementation of MOCCa stores.

Now, we change to Berts notation:

$$D^{(1)} = \begin{pmatrix} D^{(1), ul} & D^{(1), ur} \\ D^{(1), ll} & D^{(1), lr} \end{pmatrix} \tag{10}$$

where each submatrix has dimension  $N \times N$ . Let's derive Eq. (8) again. To be clear, we assume that  $i > N$ , i.e. that

$x_i > 0$  in the following equations.

$$\begin{aligned}
\frac{dh}{dx}(x_i) &= \sum_{j=1}^N D_{(i-N)j}^{(1),ll} h(x_j) + \sum_{j=1}^N D_{(i-N)j}^{(1),lr} h(x_{N+j}) \\
&= \pm \sum_{j=1}^N D_{(i-N)j}^{(1),ll} h(-x_j) + \sum_{j=1}^N D_{(i-N)j}^{(1),lr} h(x_{N+j}) \\
&= \pm \sum_{j=1}^N D_{(i-N)j}^{(1),ll} h(+x_{2N-j+1}) + \sum_{j=1}^N D_{(i-N)j}^{(1),lr} h(x_{N+j}) \\
&= \pm \sum_{\ell=1}^N D_{(i-N)(N-\ell+1)}^{(1),ll} h(x_{N+\ell}) + \sum_{j=1}^N D_{(i-N)j}^{(1),lr} h(x_{N+j}) \\
&= \pm \sum_{\ell=1}^N E_{(i-N)\ell}^{(1),ll} h(x_{N+\ell}) + \sum_{j=1}^N D_{(i-N)j}^{(1),lr} h(x_{N+j}) \\
&= \sum_{j=1}^N \left[ \pm E_{(i-N)j}^{(1),ll} + D_{(i-N)j}^{(1),lr} \right] h(x_{N+j}). \tag{11}
\end{aligned}$$

where we have (i) made the substitution  $\ell = N - j + 1$  (and hence  $j = N - \ell + 1$ ) to go from the third to the fourth line in the first sum, (ii) defined  $E_{i\ell}^{ll} = D_{i(N-\ell+1)}^{ll}$  as the ‘reversed-column-access’ to  $D$  to get the fifth line.

Comparing Eqs. (8) and (11), we see that

$$D_{ij}^{(1),\pm} = \left[ D_{ij}^{(1),lr} \pm E_{ij}^{(1),ll} \right], \tag{12}$$

where I emphasize again that  $i$  and  $j$  in this equation range from 1 to  $N$  (and not to  $2N$ ).

### III. BERT’S OUTPUT

The files Bert\_lagx/y/z.txt are written as follows

```

do i=1,nx/y/z
  do j=1,nx/y/z
    write D^{(1), +}_{ij} , D^{(1), -}_{ij}, D^{(2), +}_{ij}, D^{(2), -}_{ij}
  enddo
enddo

```

where nx/y/z of course matches  $N$  in the previous part of this document (and not  $2N$ ).

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[1] D. Baye, [TO BE FILLED].

[2] W. Ryssens et al., [TO BE FILLED].